



AWS Fundamentals

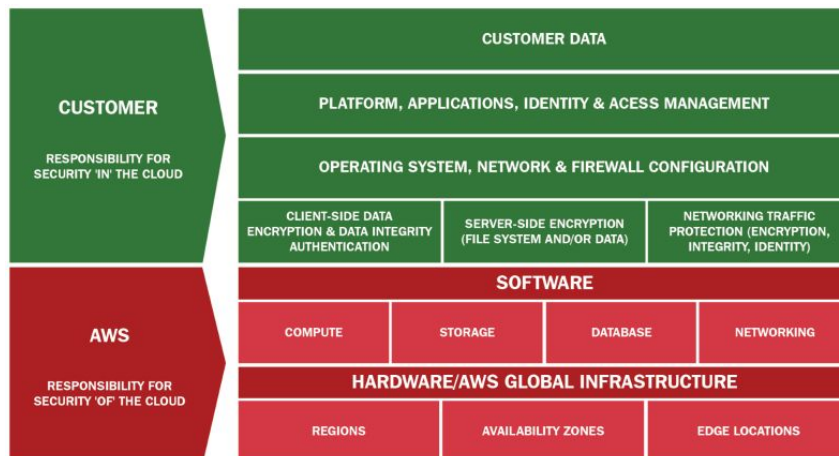
Cloud Computing

- Data Center managed by third party at the basic level
- Acquiring & Setting up infrastructure on-prem vs cloud

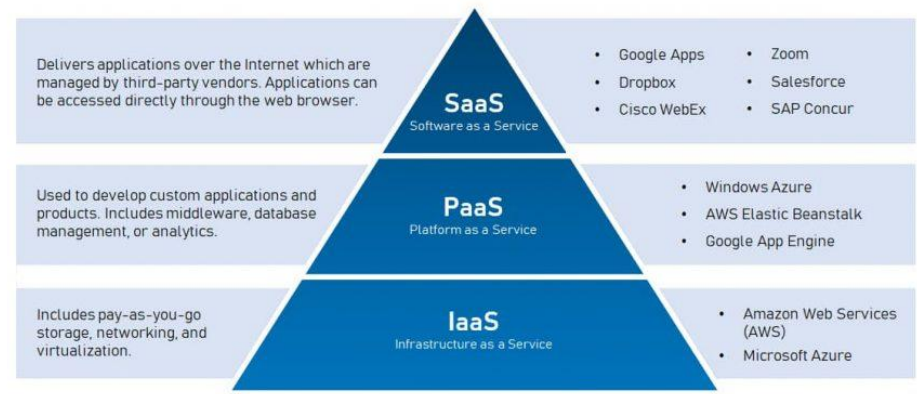
On Premise	Cloud
Plan capacity for new application	Plan capacity for new application
Check the available free capacity in the data center	
Plan for Realign of existing infra and get cost quotation for new infra	Calculate cost based on pricing available online
Submit for approval	Submit for approval
Once approved, setup infra and deploy application	Once approved, provision and deploy applications



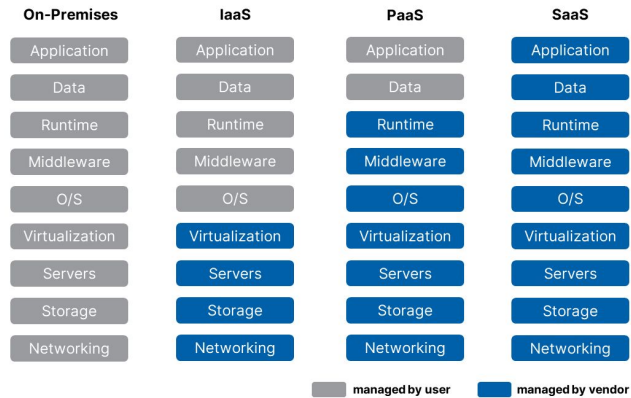
Shared Responsibility Model



IaaS, PaaS, SaaS



IaaS, PaaS, SaaS

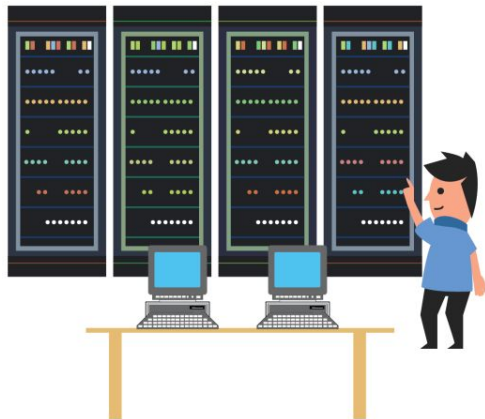


Cloud Features

- Elasticity
 - Scale up and down based on demand requirement
 - Example requirements
 - E-Commerce web store will have peak demand in holiday season
 - Payroll processing and tax filing
- Security
 - Cloud data centers are upgraded continuously with latest surveillance tech
 - Compliance standards supported by AWS
 - ITAR, SOC 1/SSAE 16/ISAE 3402, SOC 2, SOC 3, FISMA, DIACAP, and FedRAMP, PCI DSS Level 1, DOD CSM Levels 1-5, ISO 9001 / ISO 27001 / ISO 27017 / ISO 27018, MTCS Level 3, FIPS 140-2, HITRUST
- Availability
 - Zones and Regions
- Faster hardware Cycles
 - Can switch to newer processor as soon as it is made available by cloud vendor
- Other benefits
 - Capital Expenses(Capex) to Operational Expenses (Opex)
 - Faster time to market, Go global in minutes

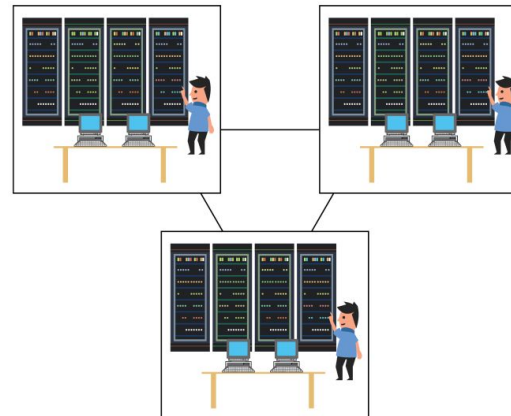


AWS Datacenter

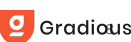
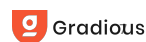


- Compute Capacity: 50,000 to 80,000 Servers
- Storage Capacity: ~ 11 petabytes per rack
- Network throughput: Upto 100 terabits per second
- Proprietary redundant network stack delivered via other device manufacturers
- Proprietary redundant network security stack

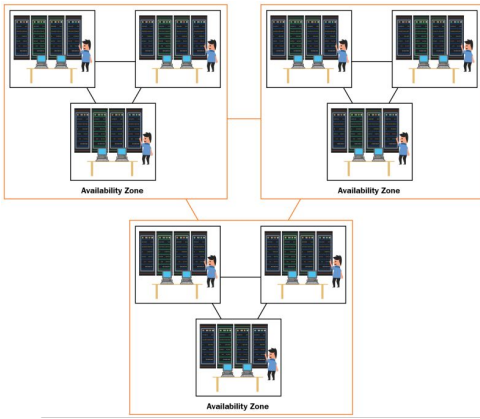
AWS Availability Zone



- Datacenters are connected together to form AZ
- Low latency layer 2 connection over metro link (2 ms)
- Geographically distributed
- Close enough to provide low latency

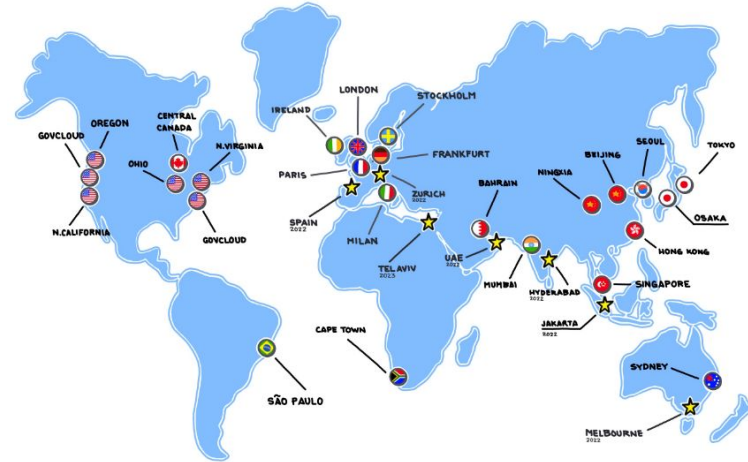


AWS Region



- Availability Zones are interconnected
- Low latency (under 5 ms)

AWS Regions Map



Choosing a Region

- Latency
- Cost
- Lack of built-in redundancy
- Recovery and Resynchronization
- Latency to clients

AWS Foundation Services

- Network Services
- Compute Services
- Storage Services
- Security and Identity Services
- End User Applications

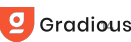
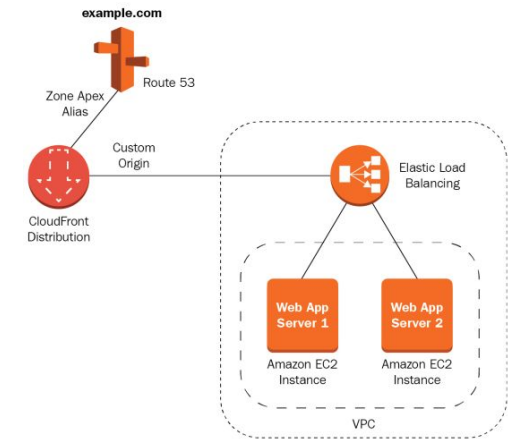
AWS Platform Services

- Databases
- Management Tools
- Analytics Tools
- Application Services
- Developer Tools
- Mobile and IoT Services



AWS Network Services

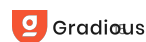
- AWS Virtual Private Cloud (VPC)
 - Network management
 - Private network
 - Publicly routable IPs
 - Connecting to internet
- AWS Direct Connect
 - Hybrid cloud with on premise services
- AWS Route 53
 - DNS as Service
 - API to use programmatically
- AWS CloudFront
 - Content Delivery Network (CDN)
 - Scalable, dynamic Caching of assets
- AWS Elastic Load Balancing (ELB)



AWS Compute Services

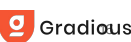
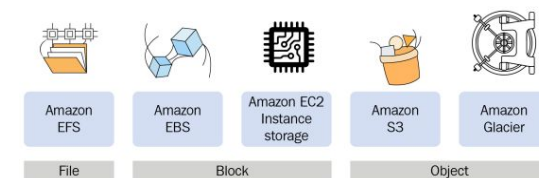
- Elastic Cloud Computing (EC2)
 - Virtual Machine Instance
 - Custom Applications
 - Variety of Operating Systems
- Elastic Container Service (ECS)
 - Containerized Applications in cloud
- AWS Lambda
 - Serverless processing of functions

Service			
	EC2	ECS	Lambda
Unit of scale	VM	Task	Function
Level of abstraction	H/W	OS	Runtime



AWS Storage Services

- Elastic Block Storage (EBS)
 - Attach block storage volumes to VMs and Containers
 - Different Sizes
 - Shared or Provisioned IOPS
- Elastic File System (EFS)
 - NFS Equivalent
- Simple Storage Service (S3)
 - Object Storage
 - Standardized & web accessible
- Amazon Glacier
 - Low cost data archiving system
- AWS Storage Gateway
 - Connect on-premise data with AWS Storage
- AWS Snow Family
 - Data transfer service
 - Move one time data from on-premise to cloud at tera and peta byte scale



AWS Security & Identity Services

- Identity and Access Management (IAM)
 - Authentication, Authorization of users and Systems
 - Supports Users, Groups, Roles & Permissions
 - Identity Federation
- Key Management Service (KMS)
 - Manage Encryption Keys
- Hardware Security Module (HSM)
 - Encryption of Data



AWS End User Applications

- Amazon WorkMail
 - Enterprise grade Email and Calendar tool
- Amazon WorkDocs
 - Cloud based Document creation and collaboration
- Amazon WorkSpaces
 - Cloud Managed Virtual Desktop Infrastructure (VDI)



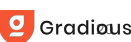
AWS Databases

- Relational Database Service (RDS)
 - Managed Database Service
 - Amazon Aurora, PostgreSQL, MySQL, MariaDB, Oracle, and Microsoft SQL Server databases
- Elastic Cache
 - Redis or Memcached in-memory data store
- DynamoDB
 - Fully managed NoSQL database
- Redshift
 - Data warehousing and business intelligence



AWS Management Tools

- Amazon CloudWatch
 - Fully managed monitoring service
 - Stores logs and environment metrics
- Amazon CloudTrail
 - Logging and monitoring of infrastructure changes
 - Logging and monitoring of API calls
- AWS Cloud Formation
 - Manage Infrastructure as code
 - Specification Template document for Infrastructure configurations
 - Deploy the infrastructure in automated repeatable manner



AWS Application Services

- Amazon API Gateway
 - Managing custom APIs at scale
 - Traffic Management, monitoring
 - Access Control and Authorization
- Amazon Elastic Transcoder
 - Cost-effective media transcoding service
- Amazon Simple Workflow Service (SWF)
 - Manage Application and Business logic workflow



AWS Developer Tools

- AWS CodeCommit
 - Git based code repository
- AWS CodeBuild
 - Continuous Integration (CI)
 - Automation of compile, run, test, deliver application package
- AWS CodeDeploy
 - Deployment of Software packages



Mobile and IoT Services

- AWS Pinpoint
 - Allows multi channel engagement. Email, SMS, Push Notifications
- AWS Device Farm
 - Cloud-enabled testing of mobile applications
- AWS Cognito
 - Single Sign-on, Authentication. Integration with Google, Facebook
- AWS Internet of Things (IoT) Services
 - Connect and manage IoT devices



AWS Networking – VPC



Knowledge Required

- Local Area Network
- Network Address Translation
- Routing
- Firewall
- IPV4
- OSI Layers



Virtual Private Cloud (VPC)

- Isolated Network in AWS Cloud
- Full control over network
- Set up own
 - Subnets
 - IP Ranges
 - Route Tables
 - Network Gateways
- Launch EC2 or ECS



VPC Subnets

- Sub network created under VPC
- Network is defined by Classless Inter-Domain Routing (CIDR) ranges
- IPV4 is used default
- IPV6 is also supported
- CIDR
 - Network part and Host part
 - Network part is denoted by / (slash)
 - CIDR of /16 – first 16 bits are used for network
 - Remaining 16 bits are used for host
 - Possible values are 0 and 1
 - Hence 2 power 16
 - /16 supports 65,536 hosts



Default VPC

- Default VPC and default subnets are created when you create account
- Use this only for PoC
- Create own VPC for production
- Default VPC configuration
 - CIDR /16 block with 172.31.0.0.
 - Default security group
 - Default network ACL
 - Default subnets in each Availability Zones with /20 range. 4096 per subnet
 - Routing enables traffic across subnets
 - Internet Gateway (IGW) is created with route for internet traffic to IGW
 - All subnets are public and public IPS are assigned automatically to all instances



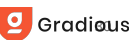
Defining Network in VPC

- Define a network at VPC level
- Break down to subnets
- Example - 1
 - 10.0.0.0/16 can be split to two /17 subnets
 - 10.0.0.0 to 10.0.127.255
 - 10.0.128.0 to 10.0.128.255
- Example 2
 - 10.0.0.0/16 can be split to four /18 subnets
 - What are the ranges?
- Network should be appropriately sized
- /16 is recommended for VPC
- Separate subnets to public and private according to service roles



Public and Private Subnets

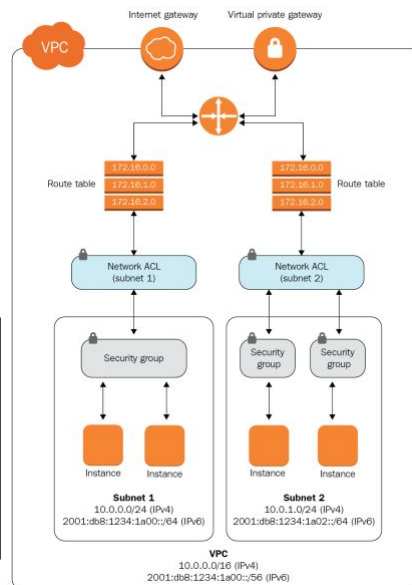
- Public subnet
 - IGW is attached
 - Route pointing to IGW is configured
 - Attach public/elastic IP to instances
 - Public IP addresses are from AWS public IP address pool
 - Elastic IP is persistent
- Private subnet
 - No IGW attached
 - No public IP address
 - Can have outbound connection with NAT gateway or NAT instance



VPC Network Security

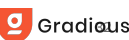
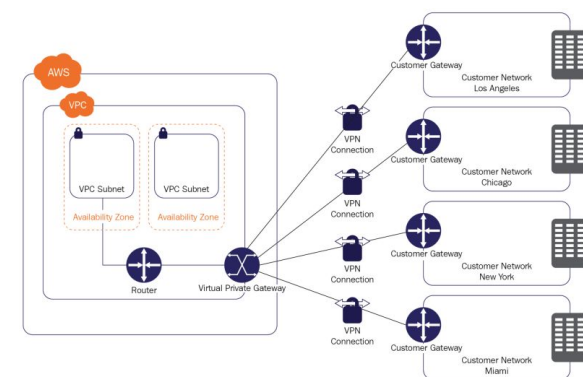
- Security Group
 - Should be applied to EC2
 - Stateful firewall
 - Port filtering
 - Can select Protocol, IP Address and port
 - Return traffic is allowed. Need not be specified
- Network ACL (NACL)
 - Stateless network control
 - Rules for one direction
 - NACL is applied to VPC and subnet
 - Used to block malicious IP ranges

Security Group	Network ACL
Operates at the instance level	Operates at the subnet level
Supports allow rules only	Supports allow rules and deny rules
Is stateful: Return traffic is automatically allowed, regardless of any rules	Is stateless: Return traffic must be explicitly allowed by rules
Evaluates all rules before deciding whether to allow traffic	Processes rules in number order when deciding whether to allow traffic
Applies to an instance only if someone specifies the security group when launching the instance, or associates the security group with the instance later on	Automatically applies to all instances in the subnets it's associated with (not subject to users specifying the security group)



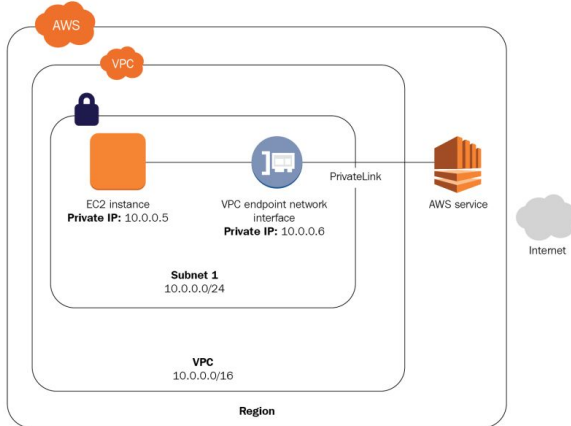
Connecting to On Premise

- VPN
 - AWS Managed VPN Gateway
 - AWS VPN CloudHub
 - VPN Instance
 - Installing VPN Server like OpenVPN
- AWS Direct Connect



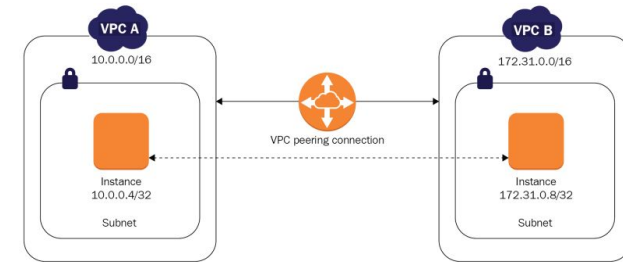
VPC Endpoint

- Connects VPC to AWS services like S3, DynamoDB



VPC Peering

- Connects VPC to other VPC
- CIDR blocks should not overlap
- Transitive peering is not supported
 - A → B and B → C does not link A and C
- IPv6 is not supported



Custom VPC

- <https://us-east-1.console.aws.amazon.com/vpc/home?region=us-east-1>

AWS EC2

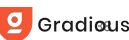
EC2 Instance Pricing Models

- On-demand Instances
- Reserved Instances
 - Standard Reserved Instances
 - Upto 75% saving over on-demand
 - Convertible Reserved Instances
 - 54% saving. Can be converted to equal or larger capacity
 - Scheduled Reserved Instances
 - Can reserve against specific schedule (fraction of day/month/year)
- Spot Instances
 - Run on spare capacity available
 - Downside: this can be terminated any time
- Dedicated Instances and Hosts
 - Dedicated Hardware. Hardware is not shared with other customers

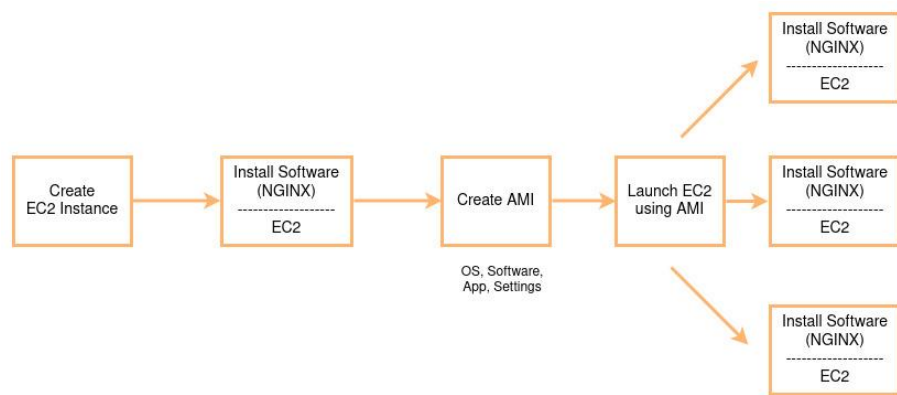


EC2 Instance Types

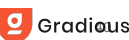
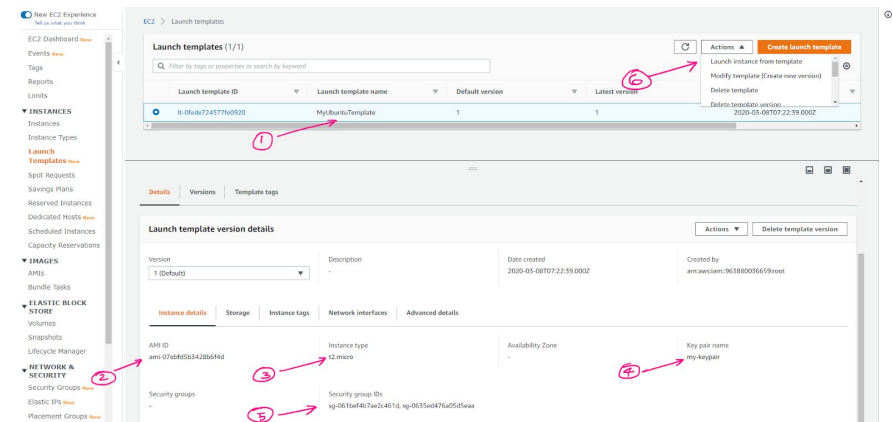
- General Purpose
 - Balance of compute, memory and networking resources
 - A type: ARM based. Good for consistent workloads
 - M type: Intel and AMD based. Good for consistent workloads
 - T type: Intel and AMD based. Cost efficient for changing workloads
 - T - Turbo burstable, M - Most scenario, A - Arm
- Compute Optimized
 - C5: Good for busy web servers and media transcoding
 - C - Compute
- Memory Optimized
 - R5: Good for in-memory databases. Real time analytics
 - X1: More demanding real time applications
 - High memory: Offers most of RAM available. Only available for 3 year reserved instances
 - X - Extra Large Memory, Z - High frequency, R - RAM
- Storage Optimized
 - H1: HDD based instances. Used for logging
 - D2: SSD based instances. Used for logging and Hadoop
 - I3: largest SSD offerings with bare metal access.
 - H - HDD, I - IO local NVMe, D - Dense
- Accelerated Computing (P3, G3, F1)
 - P3: General GPU instances. Used for machine learning
 - G3: GPU instances specialized for graphics processing
 - F1: Hardware Acceleration. FPGA (Field Programmable Gate Arrays) developer. Used for genomics, image and video processing
 - F - FPGA, G - Graphics, P - GPU



EC2 AMI



EC2 Launch Template



EC2 Store

- Root device
 - Runs Operating system
- Instance Store
 - Ephemeral block-level storage
 - Physical device attached to hypervisor
 - Ideal for temp storage, OS paging, cache, buffers
 - Data is lost if the instance is stopped or terminated
- Elastic Block Storage (EBS)
 - Persistent Storage. Highly Available
 - Survives reboots and instance failure
 - Size upto 16TB
 - EBS Volumes
 - General Purpose SSD
 - Provisioned IOPS SSD
 - Throughput Optimized HDD
 - Cold HDD
 - Ideal for infrequently accessed, throughput-intensive workloads



AWS Route 53



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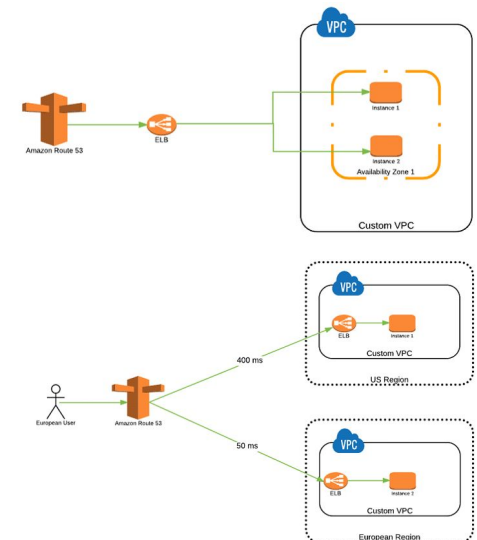
Route 53

- 100% SLA
- Classic Resolver only provides Name and IP Mapping. It does not check the availability
- Route 53 delivers address based on different parameters
 - Source and Destination Geo Location
 - Availability
 - Specific content and other rules
- Helps deliver correct content and mitigate the failure
 - Built-in health checks
 - Latency, Geo and Weightage based
 - Integration with CloudFront, S3 and ELB



Routing Policies

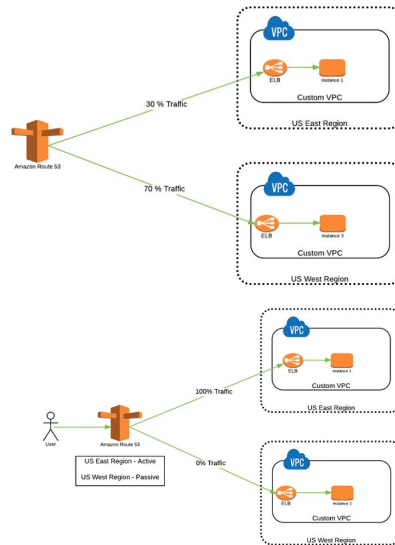
- Simple routing
 - Default. One IP address or DNS name
- Latency based routing
 - Network proximity of the user to AWS endpoint
 - Useful if you distribute the same content globally



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Routing Policies

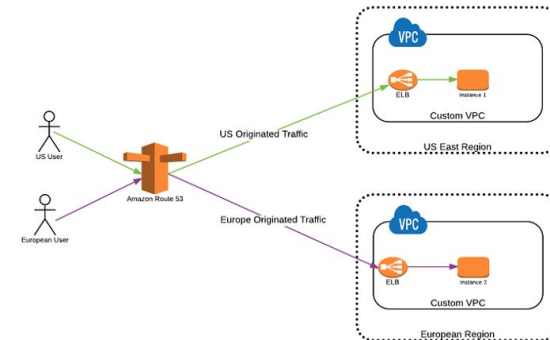
- Weighted routing
 - Can assign weight to DNS record.
 - Useful for Blue-green and A/B Testing
- Failover routing
 - Active/Passive setup



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Routing Policies

- Geolocation routing
 - Select service based on users' geo location
 - Useful for compliance too
- Multi-value response
 - Returns list of possible servers that client can choose from
- IP Based routing

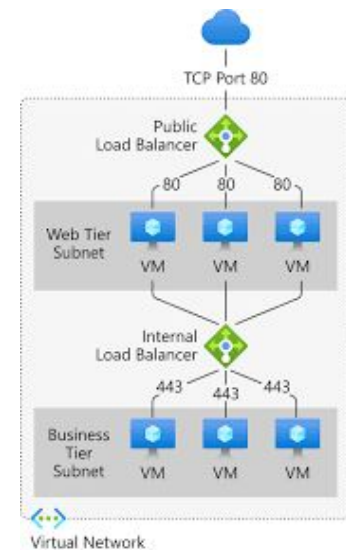


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AWS Load Balancers

Load Balancer

- Distributes the requests to different servers
- Follows routing algorithms
 - Round robin
 - Least connection
 - Request based
 - IP, Hash, Header, Query String etc
- Performs health checks and forwards the requests to only healthy servers
- Sticky Session



AWS Load Balancers

- Classic Load Balancer
 - Used to be called ELB
 - Will be retired in Aug 2022
- Application Load Balancer
 - HTTP and HTTPS traffic
 - Operates at Layer 7
 - Supports HTTP, HTTPS, HTTP/2, WebSocket
 - Advanced routing
 - Request level
 - Different target group
- Network Load Balancer
 - TCP and UDP traffic
 - Operated at Layer 4
 - High performance
 - Connection level
- Gateway Load Balancer
 - Manage Virtual Appliances such as firewalls, Intrusion detection and prevention systems and deep packet inspection systems

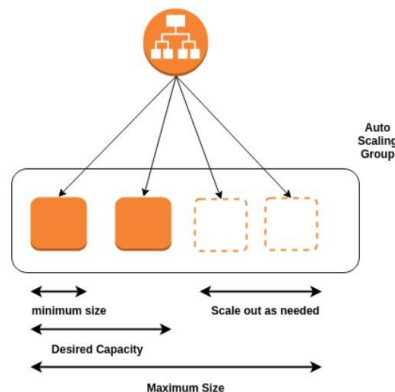


AWS Autoscaling



Autoscaling

- Auto scaling groups
- Launch configurations
- Scaling options
 - Minimum size
 - Minimum number of servers that need to be run
 - Desired Capacity
 - This will be updated whenever scale up or scale down event is fired
 - Maximum size
 - Maximum number of servers that can be run



AWS S3



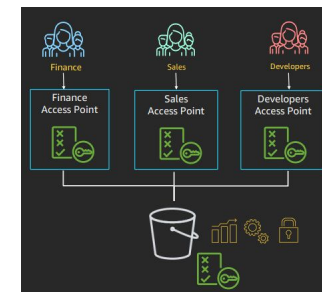
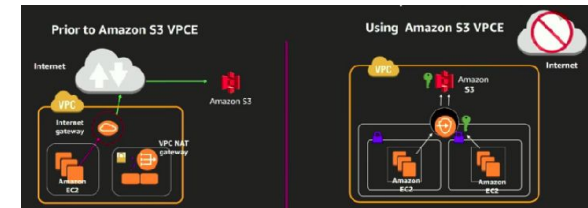
S3 – Simple Storage Service

- Fully managed, Highly available, auto scaled object/blob storage
- 99.999999999% durability 99.99% availability
- Storage Classes
 - S3 Standard
 - Replicated across three availability zones
 - S3 Intelligent Tiering
 - Moves the data to storage classes based on access frequency
 - S3 Infrequent Access
 - Lower cost per GB and higher cost per access
 - S3 One Zone–Infrequent Access
 - Replicated only within one zone. Use it for data which does not need high availability
 - Glacier
 - Instant Retrieval
 - Flexible Retrieval
 - Deep Archive
 - Low cost per GB storage. Designed for long term storage and archiving



S3 – Access

- Public Access
 - Attach to the users
 - What can this user do in AWS
- IAM Policy
 - Attach to the users
 - What can this user do in AWS
- Bucket Policy
 - Attach to the bucket
 - Who can access this S3 resource
 - Buckets are private by default
- Access Point and Policy



AWS Storage Comparison

- | EBS | S3 | EFS |
|--|---|---|
| <ul style="list-style-type: none"> • Block Storage • Data is stored in blocks • File System places data on disk (Ext4, ZFS, etc) • Must be attached to EC2 • Low latency access • Boot/data volume for EC2 • SQL and NoSQL DBs • Oracle, MySQL, Mongo DB | <ul style="list-style-type: none"> • Object Storage • Flat Structure • Prefix acts as folder • API access to data (HTTP/HTTPS GET, PUT) • Unlimited space • Media & Entertainment (images & videos) • Log Files • Backup & Archives | <ul style="list-style-type: none"> • Shared file system • Hierarchical system • Can be mounted on EC2 or on-prem • Shared access to files • Big Data & Analytics • Content Management |



AWS CLI

CLI – Command Line Interface

- Installation
 - `curl "https://awscli.amazonaws.com/awscli-exe-linux-x86_64.zip" -o "awscliv2.zip"`
 - `unzip awscliv2.zip`
 - `sudo ./aws/install`
 - `aws --version`
 - `aws configure`
 - Provide your accesskey and secret
- `aws ec2 start-instances --instance-ids i-057477c2e58321724`
- `aws ec2 stop-instances --instance-ids i-057477c2e58321724`
- `aws s3api get-object --bucket gradiousdelta --key index.html ./ind.html --region us-east-1`



AWS Cloud Formation Template



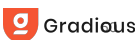
CFN – Cloud Formation Template

- Infrastructure as code
 - Infrastructure in a document
 - Document in a repository
 - Version control the document
 - Deploy the infrastructure as we deploy the code
- Objectives
 - Repeatable
 - Automated
 - Remove manual actions
 - Relieve developers, testers from mundane tasks



CFN – Steps

- Document the infrastructure
 - Feed the document to Cloud Formation Service through
 - Console
 - CLI
 - SDK
 - API
 - Cloud Formation Service deploys the resources and configures as per document
 - Rollback is possible



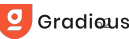
CFN – Template

- AWS Template Format Version
 - Optional. The AWS Template Format Version
- Description
 - Text that states what the template does and what services are deployed
- Metadata
 - Additional information that will be used by third party tools like designer
- Parameters
 - Values to be passed to the resources section
- Mappings
 - Environment or region specific mapping of key and values. Example: different AMI for different region
- Conditions
 - Create a resource based on a condition. Example: create autoscaling only if environment tag denotes Production
- Transform
 - To be used with serverless components like lambda. Specific version of AWS Application Model
- Resources
 - Mandatory section. Defines the actual resources that will be created by CFN service
- Outputs
 - Returns information about deployment. Example Load Balancer URL that is created during deployment



CFN – Stack

- Stack is a set of resources created by Cloud Formation Template
- Represents complete application or part of the resources from specific template
- Separate stacks to create different parts of the application (possibly managed by separate teams)
 - Network team can create a stack that contains
 - VPC, Internet Gateway, VPN
 - Security Team can create a stack that contains
 - NACL, Security Group, Users, Roles & Permissions
 - Shared Resources Team stack can contain
 - SES, SNS
 - Database Team stack can contain
 - Database and tables



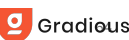
CFN – Change Sets

- Changes will be required when the stack is running
- Change Sets can be compared against the existing stack and get the report
- Three types of changes
 - Non-disruptive updates: No impact on running resources. Example: changing CloudWatch alarms.
 - Disruptive updates: Disrupts the running resource for a certain period of time. Example: Changing the instance type will require a reboot.
 - Replacement updates: Terminates an existing resource and deploys a new resource with a new ID. Example, change RDS engine from MySQL to Aurora



CFN – Stack Creation (Execution)

- Reads the template and processes them parallel
- To create resources in sequence, DependsOn needs to be used
 - Example: RDS or Database instance is required before Application EC2 gets created
- Stack Creation Status
 - CREATE_IN_PROGRESS
 - CREATE_COMPLETE
 - CREATE_FAILED



CFN – Stack Deletion

- As simple as command execution
- Deletes only if no dependant resources are there. Example:
You can not delete VPC stack if EC2 stack is dependant on the VPC stack

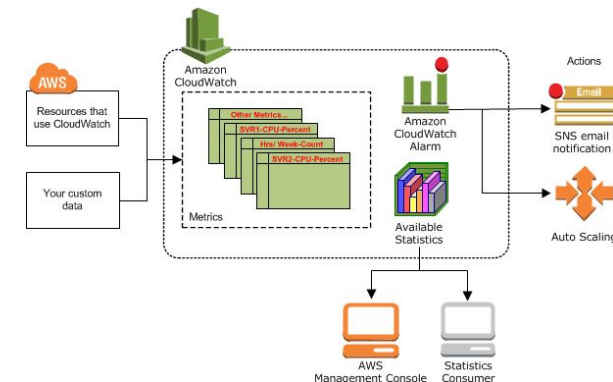
AWS CloudWatch

Monitoring

- Insights into state of the applications
 - Resource utilization graphing
 - System and application log collection
 - Resource usage metering
 - Alarming upon predefined trigger
- Many monitoring solutions are available
 - Open Source and proprietary
- Few can connect to resources and perform actions
- CloudWatch integrates with SNS, autoscaling, Lambda to perform actions

CloudWatch

- Collects metrics and logs from systems
- Every AWS service reports metrics to CloudWatch
- Supports custom metrics also
- All metrics, logs and data are stored in CloudWatch repository in specific region
- CloudWatch data is not replicated across regions
- Console, API, SDK and CLI can be used to retrieve/view the data



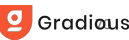
CloudWatch Elements

- Namespaces
 - Every data sent to CloudWatch should have namespace. Example AWS/EC2
 - Used to distinguish between group of metrics
- Metrics
 - Data point collected at certain point in time
 - Numeric value
 - Example: current CPU usage, Queue length of EBS volume
 - 5 minute interval by default
 - Can enable detailed metrics to collect 1 minute interval
 - Useful for alarm settings of two consequent events. 10 mins vs 2 mins
 - Data Retention
 - Below 1 minute interval data is available for 3 hours
 - 1 minute interval data is available for 15 days
 - 5 minute data is available for 63 days
 - 1 hour data is available for 15 months
 - Data is rolled up(aggregated) after expiry
- Dimensions
 - Metric is identified by dimensions
 - EC2 will have instance ID
 - Maximum of 10 dimensions per metric
 - Additional dimension gives different views. Example: Dev vs Prod
- Statistics
 - Aggregated metric collected over period of time
 - Maximum, Minimum, Sum, Average, SampleCount, pNN.NN(percentile upto 2 decimal points)
- Percentiles
 - 90th percentile means 90% of data
- CloudWatch logs
 - Application and System logs can be pushed
 - Retention period can be set
- Alarms
 - Notifications
 - Autoscaling
 - Auto-Recovery
 - Event Based Computing



CloudWatch Assignment

- Push logs to CloudWatch
- Create Alarm
- Create Dashboard



CloudTrail

- Capture
 - Capture activities recorded in AWS Services
- Store
 - Can be stored in S3, Cloudwatch Logs
- Alarm
 - Alarm can be triggered and action can be performed
- Review
 - Analyse the logs and visualize in Athena dashboard
- Useful for
 - Compliance, Auditing
 - Operational troubleshooting
 - Security analysis
 - Resource Lifecycle tracking



AWS Serverless Platform



AWS Serverless platform

- What are Serverless applications?
 - No provisioning, maintaining, and administering servers for backend components
 - Built using fully managed services as building blocks
- Why Serverless?
 - No server management
 - Flexible scaling
 - Pay for consistent throughput or execution
 - Automated high availability
- Serverless logic tier
 - Compute
 - AWS Lambda
 - Application Integration
 - Amazon API Gateway
- Data tier
 - Serverless data storage options
 - Amazon Aurora Serverless
 - Amazon RDS Proxy
 - Amazon Dynamo DB
 - Amazon S3



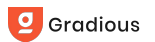
Serverless Vs EC2

Serverless

- Underlying structure managed by AWS
 - Scales automatically
 - No patching
 - Reduced risk to the cost from over-provisioning and performance from under-provisioning
- Can't install software(eg: web server, app server)in underlying environment
- Easy selection of compute power
 - 128MB to 3GB Memory
 - 1 sec to 15 min time limit
- No attached hard disk
- Inherently Highly available(Multi AZ)

EC2

- Users control underlying infrastructure –VM size, OS, AML etc..
 - Need to handle scalability
 - Requires management and orchestration
- Install almost any software
 - Prepackaged AMLs with different software available
- Adjustment of VM parameters requires some work
 - eg: may need to change ec2 instance type
- Hard Disks attached to EC2
- Need to establish HA (Multi AZ)



Serverless Vs EC2 – Use cases

Serverless

- Suitable for event driven architecture
 - Native integration with other AWS services
 - Eg: Triggered by S3, API call
- Suited when traffic is unpredictable
 - Automatic autoscaling
 - Pay as you go
- Lambda + API Gateway
- Website hosting using S3+Lambda+API Gateway
- Amazon RDS, Amazon Aurora
- DynamoDB

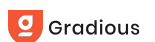
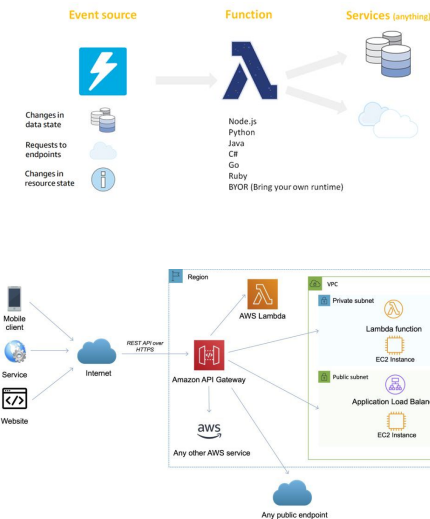
EC2

- Faster migration to cloud with other software
 - Web server, app server
 - third party software
- Suited when traffic is predicatble
 - Pay for the underlying VM
 - Scales an entire VM
- Microservices running on EC2
- Web server running on EC2
- Databases on EC2
- No SQL Databases on EC2



Serverless logic tier

- **AWS Lambda :**
 - Event driven, pay-as-you-go compute service to run code without provisioning or managing servers
- **Amazon API Gateway:**
 - Fully managed service to create and publish APIs at any scale
 - Simplified mechanisms to deploy, monitor and secure APIs
 - Improved API performance through caching and content delivery



Data tier

- **Amazon Aurora**
 - MySQL-compatible and PostgreSQL-compatible relational database built for the cloud
 - offers both serverless and traditional usage models.
 - Serverless automatically scales capacity based on application needs
- **Amazon DynamoDB**
 - key-value and document database service
 - fully managed, serverless, multi-region, multi-active, durable database
 - built-in security, backup and restore, and in-memory caching for internet-scale applications
- **Amazon RDS – Non serverless storage**
 - managed service to set up, operate, and scale a relational database using any of the available engines (Amazon Aurora, PostgreSQL, MySQL, MariaDB, Oracle, and Microsoft SQL Server)
 - optimized for memory, performance, or I/O.

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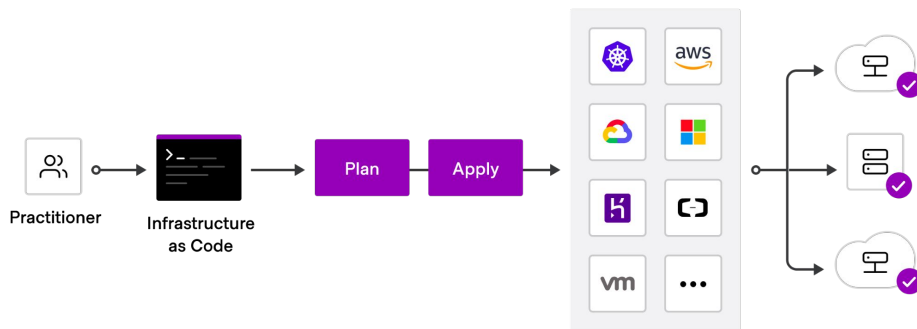
Terraform



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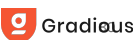
Terraform Workflow

- Init, Plan, Apply (and destroy)



Terraform components

- **Variables**
 - Key-value pair for input variables. Allows customization
- **Provider**
 - Plugin to interact with service provider API. Ex: AWS
- **Module**
 - Folder where all terraform templates are defined
- **State**
 - Cached information about infrastructure managed by Terraform
- **Resources**
 - Block for infrastructure objects
- **Data Source**
 - Implemented by providers to return information on external objects to Terraform
- **Output Values**
 - Return values of a terraform module that can be used by other configurations
- **Plan**
 - One of the stages that determines what needs to be created, updated, or destroyed
- **Apply**
 - One of the stages that applies the change



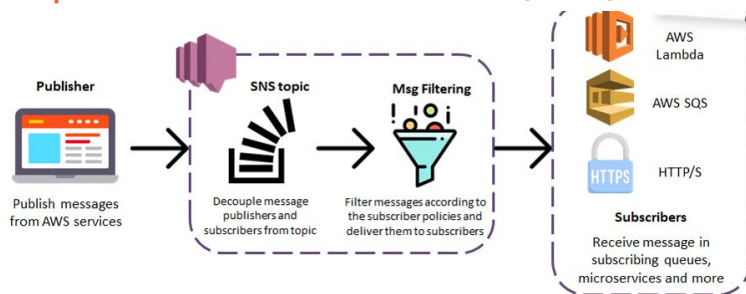
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Messaging Platforms

Messaging Platforms

- **Amazon Simple Notification Service (SNS)**
 - fully-managed pub/sub messaging system
 - to deliver messages to large numbers of end users, such as via mobile push notifications or SMS messages.
- **Amazon Simple Queue Service (SQS)**
 - fully-managed queue service and offers flexible queuing system
 - used to offload long-running tasks in web applications or to communicate between microservices
 - SQS uses HTTP rather than a stateful connection to the message broker.
- **Amazon Simple Email Service (SES)**
 - fully-managed email service for both sending and receiving email.
 - main use case for SES is sending email (transactional & marketing)
 - Use SES to receive emails for automating responses or handling inbound ticket requests.

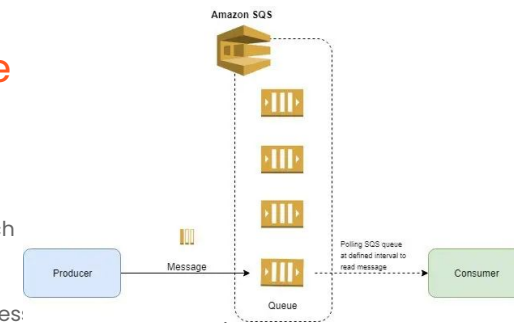
Simple Notification Service (SNS)



- push-based, many-to-many messaging service
- Benefits
 - Reliability
 - Message Filtering
 - Automatic Scalability
 - Secure

Simple Queue Service

- Uses HTTP as the protocol for interacting with SQS.
- Standard queue :
 - no guarantees about the order in which messages will be processed
- First-in, first-out (FIFO) queues :
 - guarantee that messages will be processed
 - useful for non-idempotent tasks where subsequent messages depend on the results from prior messages.
- Long-polling
 - When queue is empty, need to continually ping the queue to see if any new messages have arrived
- Dead-letter queues (DLQ):
 - sends messages that has been unsuccessfully processed a given number of times.



- 

```

graph LR
    User((User)) -- purchase --> TPWS[Transaction Processing Webservice]
    TPWS --> CCA[CreditCard Authority Service]
    TPWS --> SNS[Transaction SNS Topic]
    SNS --> CRS[Customer Reminder Service]
    SNS --> TAQ[Transaction Analytics Queue]
    SNS --> TFQ[Transaction Fraud Detection Queue]
    CRS -- "Thanks for ordering!" --> CE[Customer Email]
    TAQ -- poll --> TAS[Transaction Analytics Service EC2]
    TFQ -- poll --> TFD[Transaction Fraud Detection]
    TAS -- "OrdersToday: 225, TotalRevenue: 12000" --> AnalyticsOut[Analytics Out]
    TFD -- "Suspect Transactions: [\"758191\", \"262133\"]" --> FraudOut[Fraud Out]
  
```

The diagram illustrates a serverless architecture for a payment system. It starts with a user initiating a 'purchase' through a 'Transaction Processing Webservice'. This service then interacts with a 'CreditCard Authority Service' and publishes a message to a 'Transaction SNS Topic'. The SNS Topic triggers three parallel services: 'Customer Reminder Service' (which sends a 'Thanks for ordering!' email), 'Transaction Analytics Queue' (which is polled by 'Transaction Analytics Service (EC2)' to get 'OrdersToday: 225, TotalRevenue: 12000'), and 'Transaction Fraud Detection Queue' (which is polled by 'Transaction Fraud Detection' to get 'Suspect Transactions: [\"758191\", \"262133\"]').

- Highly scalable, low-latency content delivery network
- Delivers
 - Static content
 - Cacheable dynamic content
 - Dynamic data, videos, images
- Lowers the latency and delivery time
- 300+ Edge locations (90+ cities and 47 countries)
- Uses highly resilient private backbone network
- Regional edge caches
 - Even after edge cache expiry, data can be served from regional edge cache

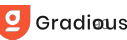
CloudFront Features

- Origin can be
 - S3
 - Any DSN resolvable HTTP/HTTPS server
 - Hosted in AWS or outside
- Security
 - Transit encryption (HTTPS/TLS)
 - Can use AWS Certificate Manager to get free certificate and integrate
- Restriction
 - Origin Access Identity
 - By geography
 - Signed URL or cookie



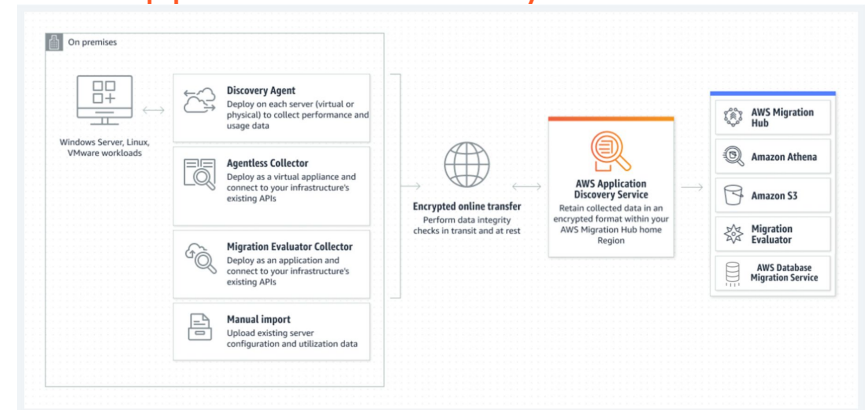
Content Delivery Methods

- GET, HEAD, POST, PUT, OPTIONS, DELETE, PATCH
- Cache Behavior
 - Time to Live (TTL)
 - MIN
 - If no-cache, no-store, private are present, MIN will be used
 - Default
 - Used when the cache control headers do not provide a caching duration
 - MAX
 - Overrides the cache control headers by enforcing a shorter caching duration



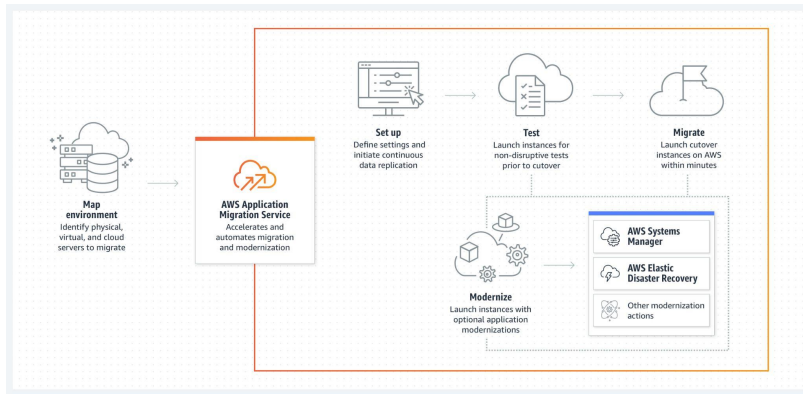
AWS Migration & Transfer

AWS Application Discovery Service



- Discover on-premises infrastructure
- Identify server dependencies
- Measure server performance
- Data Exploration in Amazon Athena

AWS Application Migration Service



- CloudEndure Agent should be installed in each server
- Machines will be shown in CloudEndure console
- Can migration applications and databases



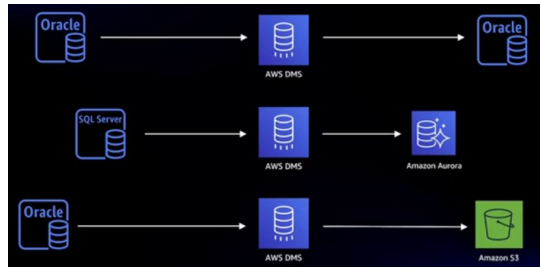
DMS – Database Migration Service

- Migrates and/or replicates databases and data warehouses to AWS
- 500,000 databases have been migrated as of 2021
- Requirement to use DMS
 - Hardware end of life
 - Lack of redundancy
- Scalability
 - Can change instance sizes as required
 - Scale across regions as required
- Cost
 - Fully managed databases
 - Pay for what you use

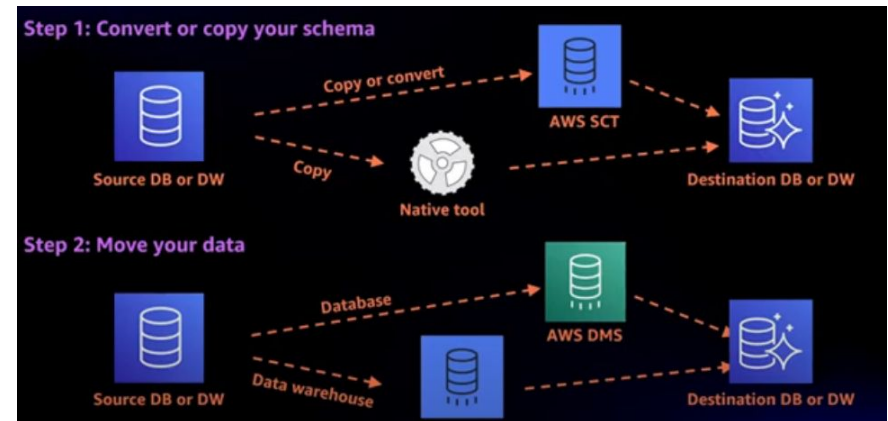


SCT – Schema Conversion Tool

- Homogeneous database migration
 - Same source and target databases
 - MySQL to MySQL RDS or Oracle to Oracle RDS
- Heterogeneous database migration
 - Different source and target databases
 - MySQL to Oracle RDS or Oracle to MySQL RDS
- Can convert Secondary objects like Stored Procedure and Functions
- Creates assessment reports for both homogeneous and heterogeneous database migration
- Schema
 - Convert database schema
 - Convert data warehouse schema

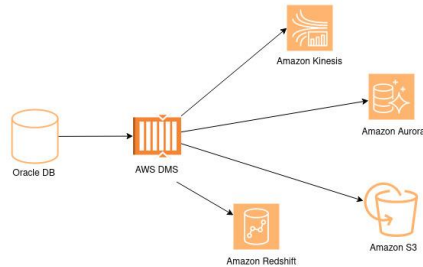


Data Migration



Data Migration

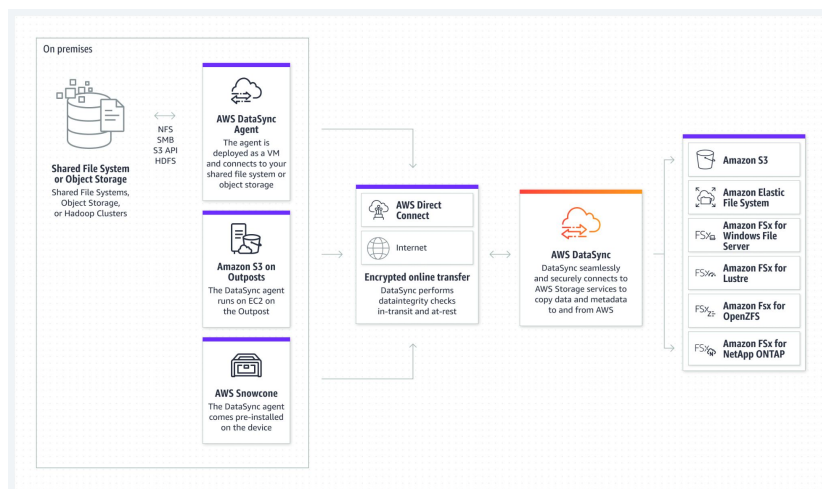
- Process
 - Assess, Plan, Convert Schema and code, Optimize, Migrate
- Migration
 - Data Migration from Source to Destination
- Data Replication
 - Continuous Data Capture (CDC)
 - Reads from transaction logs
 - Streams data changes to target database
- Features
 - Secure
 - Assessment
 - Validation
 - Monitor
 - Multiple Option
 - Fanout
 - From single source to multiple source



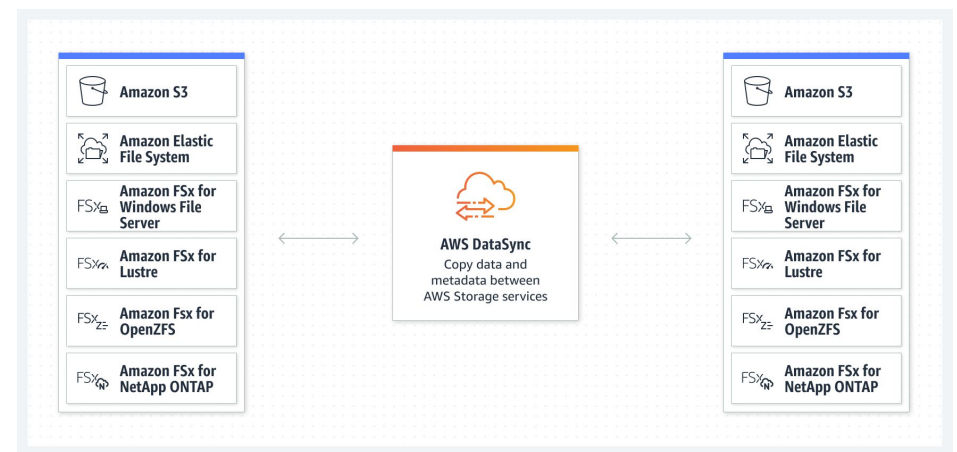
Data Migration – General Use Cases

- Migration
 - Data Migration
 - Migrate SQL to SQL, SQL to NoSQL, NoSQL to SQL, NoSQL to NoSQL
 - Upgrade to different major/minor version
 - Archive old data to S3
- Replication
 - Cross region read replicas
 - Analytics in the cloud

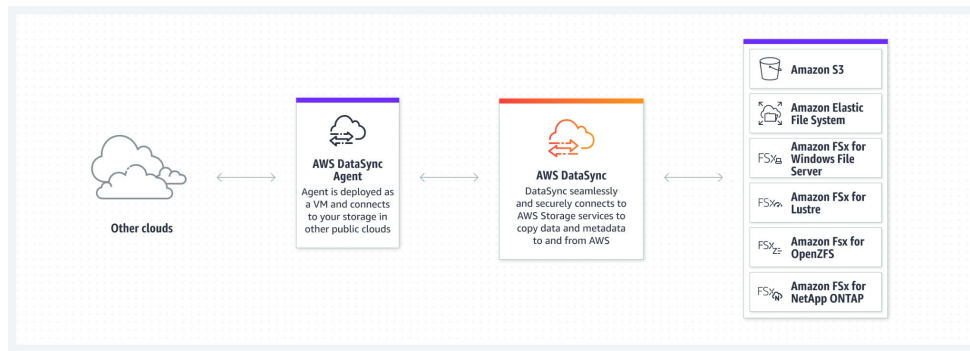
DataSync – Transfer Between On Premise and AWS



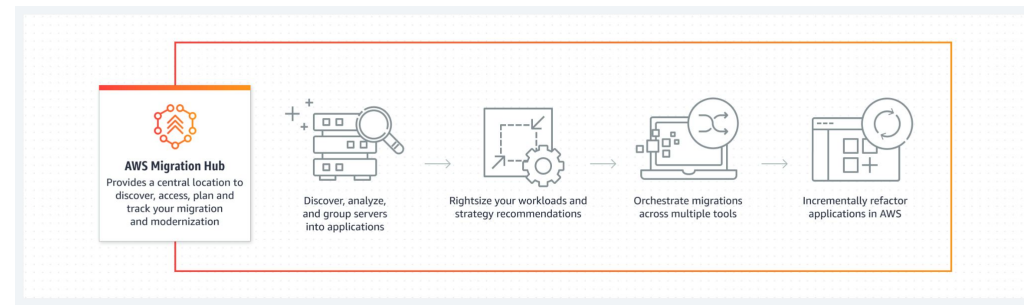
DataSync – Transfer Between AWS Storage Services



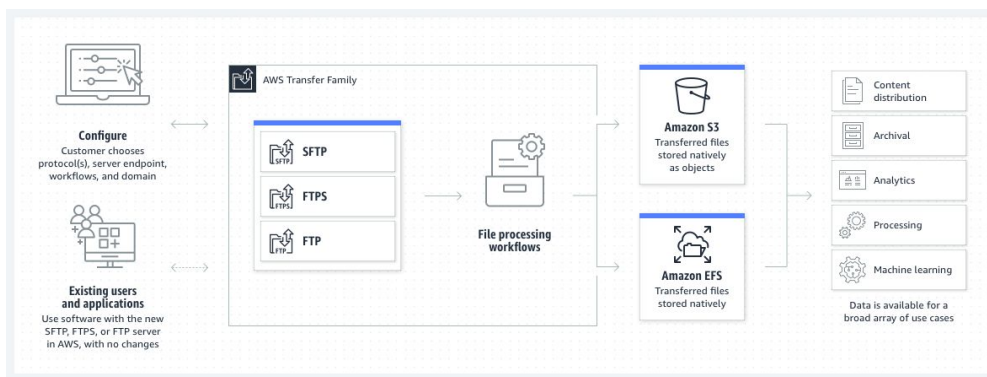
DataSync - Transfer Between AWS and Other Clouds



AWS Migration Hub



AWS Transfer Family

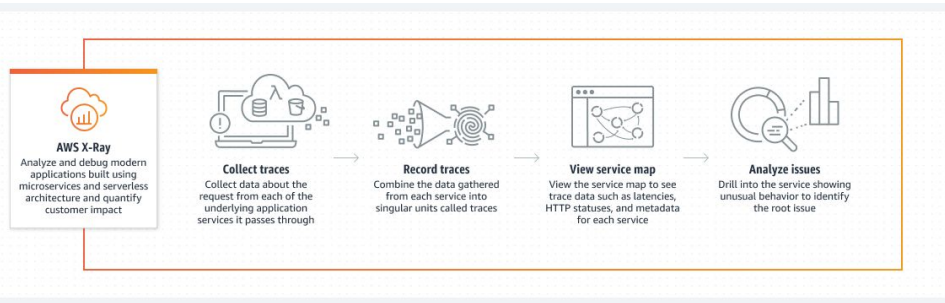


AWS Snow Family



X-Ray

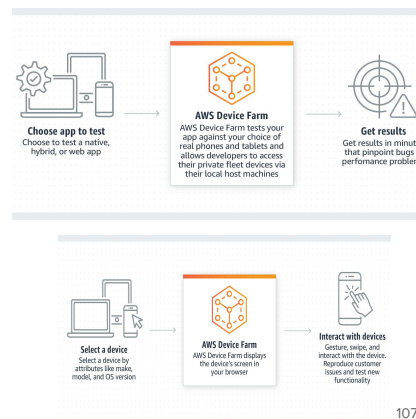
- To conduct Perform analysis and debug distributed applications
- Provides a complete view of requests as they travel through the application and filters visual data across payloads, functions, traces, services, APIs, and more



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Device Farm

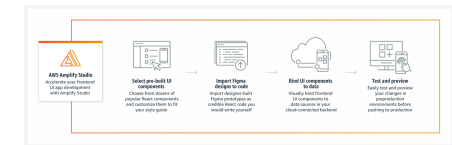
- Improve the quality of the web and mobile applications by testing across desktop browsers and real mobile devices hosted in the AWS Cloud
- Runs the tests concurrently on multiple desktop browsers or real devices to speed up the execution of the test suite
- Generates videos and logs to help you quickly identify issues with your app.
- Provides the benefits of real device testing by setting up private device lab in the cloud and simulating real-world environments



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Amplify

- Build feature-rich, full-stack web and mobile apps from development to deployment.
- Create an AWS backend for web, iOS, or Android app with authentication, data, storage, and more
- Visually build a frontend UI with design-to-code Figma integration and connect the UI to a backend in clicks.
- Deploy and host fast, secure, reliable websites and server-side rendered apps in a few clicks

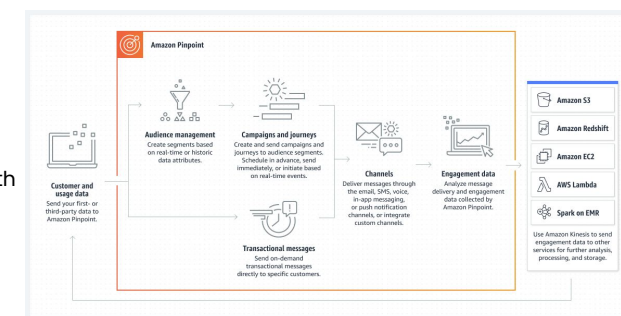


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Pinpoint

- Connect with customers through scalable, targeted multichannel communications

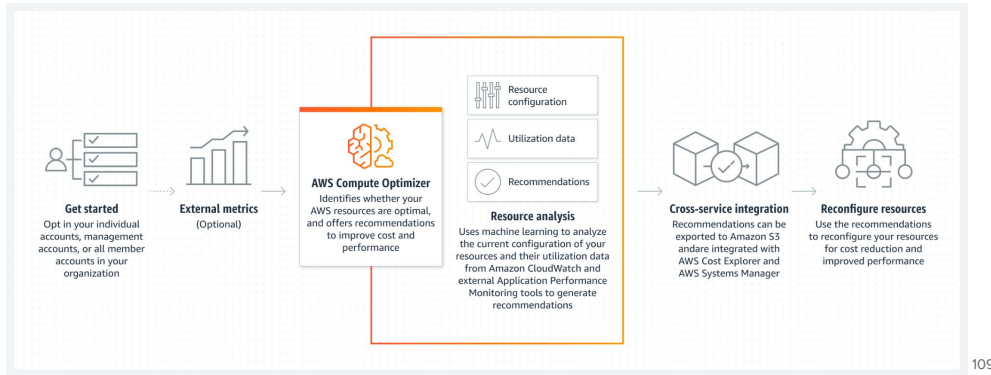
- Define audience segments
- Engage your audience with messaging campaigns
- Create user journeys
- Provide consistent messaging with templates
- Deliver personalized content
- Analyze user behavior



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Compute Optimizer

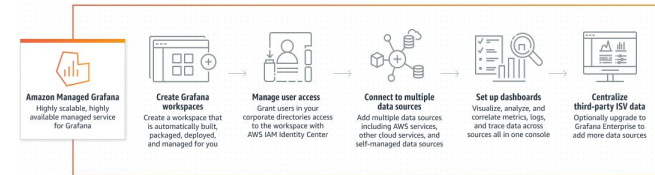
- Analyzes the configuration and utilization metrics of AWS resources (EC2 instances, EC2 Auto Scaling groups, EBS volumes, ECS services on AWS Fargate, and Lambda functions.)
- Generates optimization recommendations to reduce the cost and improve the performance of the workloads



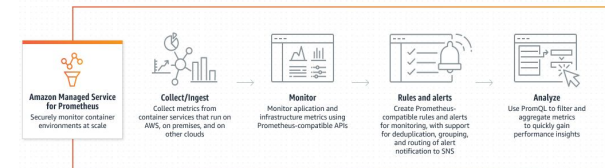
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Managed Grafana & Prometheus

- Amazon Managed Grafana: Fully managed and secure data visualization service to instantly query, correlate, and visualize operational metrics, logs, and traces from multiple sources.



- Amazon Managed Service for Prometheus: Serverless, Prometheus-compatible monitoring service for container metrics that makes it easier to securely monitor container environments at scale.



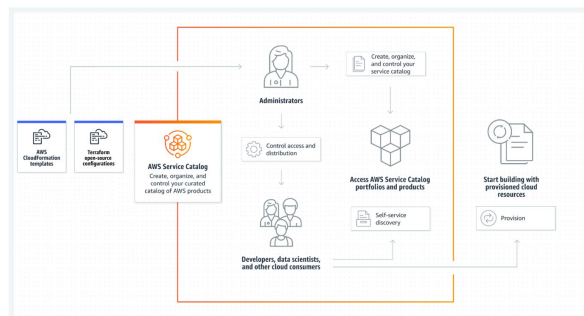
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Organizations & Service Catalog

- AWS Organizations: Consolidate multiple AWS accounts for centralized management of budgetary, security, and compliance needs of the business



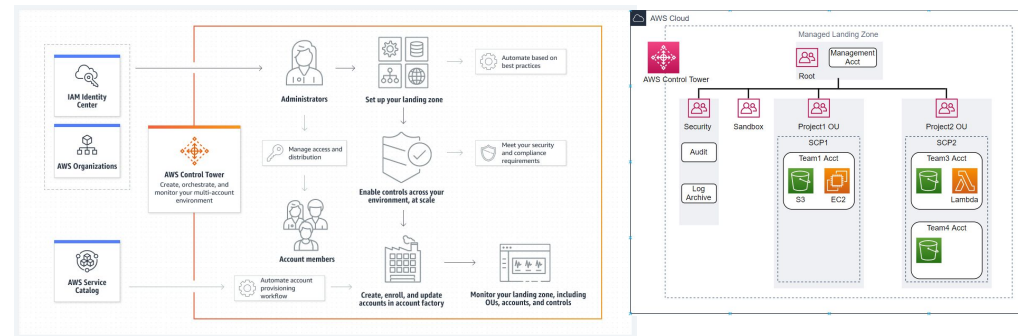
- AWS Service Catalog: Centrally manage the cloud resources to achieve governance at scale of the infrastructure as code (IaC) templates, written in CloudFormation or Terraform.



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Control Tower

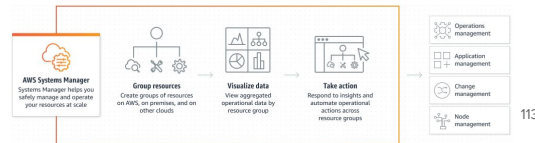
- Provision, set up and govern a secure, multi-account AWS environment, known as a landing zone
- Combines the abilities of AWS Organisations, AWS Service Catalogue, and IAM Identity Centre



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Config, License & Systems Manager

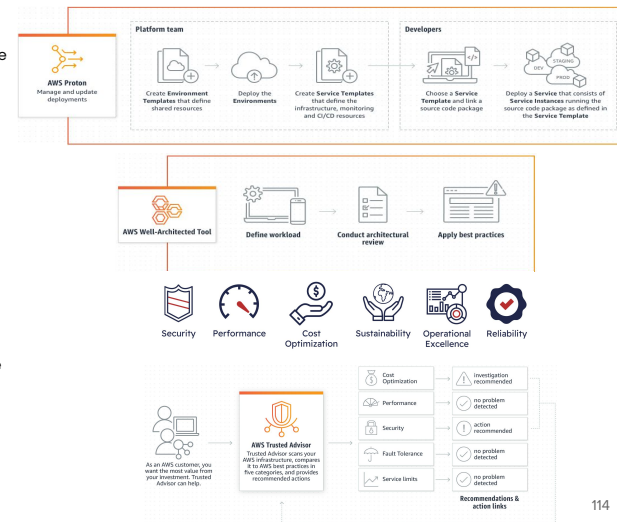
- **AWS Config**
 - Continually assess, audit, and evaluate the configurations and relationships of resources on AWS, on premises, and on other clouds
- **AWS License Manager**
 - Manage software licenses from vendors, such as Microsoft, SAP, Oracle, and IBM, across AWS and on-premises environments.
- **AWS Systems Manager**
 - Operations hub for AWS applications and resources providing secure end-to-end management solution



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Few other Services...

- **AWS Proton**: Fully managed delivery service for container and serverless applications to connect and coordinate all the different tools needed for infrastructure provisioning, code deployments, monitoring, and updates.
- **AWS Well-Architected Tool** : Get architectural guidance, access to best practices, and an action plan to fine-tune applications and workloads.
- **AWS Trusted Advisor**: To analyze the AWS environment and recommend actions to follow best practices.
- **AWS Personal Health Dashboard**: Provides a personalized view of the AWS service status that powers the application, learn about specific operational issues that affect the account.



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